Michael Sobrepera

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Summary

Robotics researcher, systems engineer, and business/product leader with deep expertise in mechanical design, perception-to-control integration, hardware/software system integration, AI-enabled robotic systems, and human robot interaction.

- 7+ years developing complex robotic systems from concept through deployment, including full-stack development of an upper-humanoid social robot (mechanical design, control, sensor integration, computer vision, social human interaction, and system-level integration) at the University of Pennsylvania and vision-based tracking and control systems for industrial manufacturing at The Georgia Institute of Technology.
- 3+ years driving AI strategy and product development at the Boston Consulting Group (BCG), with hands-on experience building and deploying ML systems and leading cross-functional teams.
- Proven record of leading teams, owning products, developing strategies, and managing complex stakeholder interactions.

Seeking to apply deep technical expertise on cutting-edge robotics challenges.

Education

Doctor of Philosophy in Mechanical Engineering Aug 2016 – Apr 2022
University of Pennsylvania *Philadelphia, PA*
Advisor: Dr. Michelle Johnson
Affiliated with the General Robotics, Automation, Sensing, & Perception Laboratory (GRASP Lab)

Master of Science in Robotics Aug 2016 – Aug 2019
University of Pennsylvania *Philadelphia, PA*

Bachelor of Science in Biomedical Engineering Aug 2012 – Dec 2015
Georgia Institute of Technology *Atlanta, GA*
Minor in Computer Science

Experience

Project Leader Aug 2024 – present
Consultant Apr 2022 – Aug 2024
Boston Consulting Group (BCG) *Seattle, WA*

- Led 8-person cross-functional team driving product, go-to-market, and stakeholder management for an agentic AI product; accelerated user testing, product iteration, and adoption to maximize value.
- Led 3-person team building BCG's partnership with a hyperscaler to mature a pipeline of AI and digital build opportunities, including acting as the expert to help BCG teams define the capability/product to be built.
- Guided 10+ client organizations to define an AI vision and AI strategy, through workshops with senior leadership, and to develop AI roadmaps and build plans, in coordination with technical leadership, enabling clients to rapidly move toward value capture from AI.
- Built MVP AI products enabling non-technical users to query heterogeneous survey data to improve strategic decision making and analyze incident reports to respond to infrastructure failures.
- Redesigned and implemented the leadership management system and core KPIs of a 15k+ person industrial facility to deliver a turnaround which leadership described as the most impactful shift in a generation.
- Built a large-scale data processing pipeline (in R) to target a multi-million dollar global health investment with a clear focus on need, providing data to a large team of business consultants and client data scientists.

NIH F31 Fellow Apr 2020 – Apr 2022
Graduate Research Assistant Aug 2016 – Apr 2022
University of Pennsylvania, Rehabilitation Robotics Laboratory *Philadelphia, PA*

- Led complete system design and integration of an affordable social rehab robot (upper-humanoid) for telerehabilitation: designed mechanical structure and actuator integration, developed motor control and feedback algorithms, integrated sensors (incl. RGBD cameras), performed full system integration, developed teleop and semi-autonomy capabilities, developed human/robot behavior system, and drove testing (`https://youtu.be/DDZe1RhcpWY`, `https://github.com/Rehab-Robotics-Lab/FloSystem`).
- Studied patient interactions with telepresence systems which incorporate social robots and impacts on remote assessment and treatment after stroke, cerebral palsy, etc.
- Developed a prototype framework for measuring motor function from video of patients engaging in robot-guided activities using both classical computer vision and machine learning techniques.

Research Technician II Dec 2015 – Jun 2016
Undergraduate Research Assistant Aug 2015 – Dec 2015
Georgia Institute of Technology, IRIM Technology Transition Laboratory *Atlanta, GA*

- Translated a vision-based 3D object tracker from a research code base to a robust system capable of running reliably, in real-time, and at high resolution to enable robotic manipulation on non-fixed, non-located parts in manufacturing.
- Developed and validated perception and control systems for industrial manufacturing applications, working with heavy industrial robots and optimizing for real-time performance and reliability in production environments.
- Developed tools for calibrating multiple robot arms to cameras.
- Integrated perception and motion control to track moving targets with collaborative robots using vision guidance.

Automation Intern May 2015 – Aug 2015
Eli Lilly and Company *Indianapolis, IN*

- Developed strategy for offline software/hardware/operator-in-the-loop plant simulations (digital twins) for process validation, control code development, control system checkout, operator training, and process improvement.

Machine Shop Supervisor Aug 2014 – May 2015
Georgia Institute of Technology, TEP Machine Shop *Atlanta, GA*

- Maintained shop, ensuring availability and safety of equipment.
- Guided Master's in Biomedical Innovation and Development students in designing and prototyping of medical devices.
- Supported cardiovascular research labs in development and fabrication of experimental equipment.

Product Development Engineering Co-Op Jan 2014 – Jul 2014
Unilife Corporation *York/King of Prussia, PA*

- Developed, prototyped, and tested new injectable drug delivery device product concepts.
- Designed and integrated automation equipment for syringe component gluing, assembly, and finish operations.

Highlighted Skills

Robotics Engineering	HRI, System and Sensor Integration, ROS, Computer Vision, Planning & Control
Software Engineering	Python, C++, R, Docker, Git, ML/AI Integration
Mechanical Engineering	Product Design, Mechatronics, MCAD (SolidWorks), Prototype and Scale Manufacturing
Product & Strategy	Product Discovery, Roadmapping, Feature Definition, Go-To-Market, Stakeholder Management

Publications

Peer-reviewed Journal Publications

[1] **Michael J. Sobrepera**, Anh T. Nguyen, Ajay Anand, Laura A. Prosser, Sally H. Evans, and Michelle J. Johnson. "Age, Motor Function, and Cognitive Function Influence Preferences for Telerehabilitation Mediated by a Social Robot Augmented with Telepresence". In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2025). DOI: 10/pzqz.

[2] **Michael J. Sobrepera**, Anh T. Nguyen, Emily S. Gavin, and Michelle J. Johnson. "Insights into the deployment of a social robot-augmented telepresence robot in an elder care clinic – perspectives from patients and therapists: a pilot study". In: *Robotica* (2024), pp. 1–29. DOI: 10/mpv6.

[3] **Michael J. Sobrepera**, Julie Elfishawy, Anh T. Nguyen, Laura P. Prosser, and Michelle J. Johnson. "Insights on Telecommunication Use by Rehabilitation Therapists Before, During, and Beyond COVID-19". In: *Archives of Rehabilitation Research and Clinical Translation* (2024), p. 100326. ISSN: 2590-1095. DOI: 10/mpv7.

[4] **Michael J Sobrepera**, Vera G Lee, and Michelle J Johnson. "The Design of Lil'Flo, a Socially Assistive Robot for Upper Extremity Motor Assessment and Rehabilitation in the Community via Telepresence". In: *Journal of Rehabilitation and Assistive Technologies Engineering* 8 (Apr. 19, 2021), pp. 1–26. ISSN: 2055-6683. DOI: 10/gjrpj8.

[5] **Michael J Sobrepera**, Vera G Lee, Suveer Garg, Rochelle Mendonca, and Michelle J Johnson. "Perceived Usefulness of a Social Robot Augmented Telehealth Platform by Therapists in the United States". In: *IEEE Robotics and Automation Letters* 6.2 (2021), pp. 2946–2953. ISSN: 2377-3766. DOI: 10/gh6hkk.

[6] Michelle J Johnson, **Michael J Sobrepera**, Enri Kina, and Rochelle Mendonca. "Design of an Affordable Socially Assistive Robot for Remote Health and Function Monitoring and Prognostication". In: *International Journal of Prognostics and Health Management (IJPHM)* 10. Special Issue PHM for Human Health and Performance (2019).

Peer-reviewed Conference Publications

[1] **Michael J. Sobrepera**, Vera G Lee, and Michelle J. Johnson. "Therapists' Opinions on Telehealth, Robotics, and Socially Assistive Robot-Augmented Telepresence Systems for Rehabilitation". In: *2022 9th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*. Seoul, Korea, Republic of: IEEE Press, 2022, pp. 1–8. DOI: 10/mfnn.

[2] **Michael J. Sobrepera**, Vera G. Lee, Suveer Garg, and Michelle J. Johnson. "Feasibility and Acceptability of Remote Neuromotor Rehabilitation Interactions Using Social Robot Augmented Telepresence: A Case Study". In: *2022 International Conference on Rehabilitation Robotics (ICORR)*. 2022, pp. 1–6. DOI: 10/gsgmmk.

[3] **Michael J Sobrepera**, Enri Kina, and Michelle J Johnson. "Designing and Evaluating the Face of Lil'Flo: An Affordable Social Rehabilitation Robot". In: *IEEE International Conference on Rehabilitation Robotics*. Toronto, Ontario, Canada, 2019. DOI: 10/ggdd87.